



East West University
Department of Computer Science and Engineering
Course: CSE103 (Structured Programming) LAB

LAB 04: Arrays and Multidimensional Arrays

Lab Objectives

- Understand the concept and declaration of arrays in C.
- Learn how to store and access multiple data elements using a single variable name.
- Apply 1D arrays to perform mathematical and statistical operations.
- Use 2D arrays (matrices) to perform tabular computations.
- Strengthen logic through real-world examples involving lists, scores, and grids.

Topics Covered

- 1D Array declaration and initialization
- Accessing and updating array elements
- Traversing arrays using loops
- Multidimensional arrays (2D)
- Nested loops for 2D array input/output
- Basic operations: sum, average, search, max/min, matrix addition

Lab Outcomes

After successful completion of this lab, students will be able to:

- Declare and manipulate arrays effectively.
- Store and process multiple data values efficiently.
- Apply nested loops to read and display 2D array data.
- Perform computations such as totals, averages, and comparisons using arrays.

Lab Activities

Part A: One-Dimensional Array (1D)

Problem 1: Input and Display Array Elements

Write a C program to take n integers as input and display them in reverse order.

Input:	Output:
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5 10 20 30 40 50	50 40 30 20 10
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Problem 2: Find Maximum and Minimum Element from an array

Write a C program that takes n numbers as input in an array and displays the maximum and minimum element of the array.

Input: 5 10 3 56 78 2	Output: Max = 78 Min = 2
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Problem 3: Count Even and Odd Numbers

Write a C program that takes n numbers as input in an array and displays the number of even and odd numbers in it.

Input: 5 3 5 6 8 9	Output: Even = 2, Odd = 3
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Problem 4: Student Grade Analysis

Write a C program to read marks of n students (0 – 100) and determine:

- The highest, lowest, and average marks.
- The number of students who passed (marks ≥ 40).

Input: 5 45 72 39 80 60	Output: Highest = 80 Lowest = 39 Average = 59.20 Passed Students = 4
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Problem 5: Reverse the Array

Write a C program that takes n integers as input and prints only the even numbers in reverse order.

Input: 5 1 2 3 4 6	Output: 6 4 2
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Problem 6: Search an Element in Array

Write a C program that takes an array of n numbers and a value x , and checks whether x exists in the array or not.

Input: 5 2 4 6 8 10 Enter value to search: 6	Output: 6 found at position 3
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Problem 7: Daily Expense Tracker

You record your spending for n days in an array. Write a program to calculate total expense and day with maximum and minimum expense

Input: 7 200 150 400 120 350 100 180	Output: Total = 1500 Highest Expense = 400 on Day 3 Lowest Expense = 100 on Day 6
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Problem 8: Vote Counting System

In an election, there are m candidates. Each vote is stored as a candidate number ($1-m$) in an array of size n . Write a program to count the number of votes each candidate received and find the winner.

Input: Number of candidates: 3 Number of votes: 10 Votes: 1 2 1 3 2 1 1 3 2 2	Output: Candidate 1 received 4 votes Candidate 2 received 4 votes Candidate 3 received 2 votes Result: Tie between Candidate 1 and Candidate 2
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Part B: Two-Dimensional Array (2D)**Problem 9: Input and Display a Matrix**

Write a C program to read and display a 2D array (matrix) of size $m \times n$.

Input: 2 3 1 2 3 4 5 6	Output: 1 2 3 4 5 6
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Problem 10: Sum of Diagonal Elements

Write a C program that takes a square matrix ($n \times n$) as input and prints the sum of the primary diagonal values.

Input: 3 1 2 3 4 5 6 7 8 9	Output: Sum of main diagonal = 15
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Problem 11: Transpose of a Matrix

Write a C program that takes a $n \times m$ matrix as input and displays the transpose of a matrix.

Input:	Output:
2 3	1 4
1 2 3	2 5
4 5 6	3 6

Problem 12: Upper and Lower Triangular Matrix (Square Matrix Only)

Write a program that takes a $n \times n$ square matrix as input and prints the upper and lower triangles of a square matrix.

Input:	Output:
1 2 3	upper triangle
4 5 6	1 2 3
7 8 9	5 6
	9
	Lower triangle
	1
	4 5
	7 8 9

Problem 13: City Temperature Analyzer

You are given the temperature data of a city for 7 days (rows) at 4 different times each day (columns).

Write a C program that:

1. Calculates and prints the average temperature per day.
2. Displays which day had the highest average temperature.

Input: (Temperature Data in °C):	Output:
Day 1: 30 32 31 29	Average temperature per day:
Day 2: 28 30 29 27	Day 1 average 30.50°C
Day 3: 33 34 35 32	Day 2 average 28.50°C
Day 4: 26 28 30 31	Day 3 average 33.50°C
Day 5: 29 27 28 26	Day 4 average 28.75°C
Day 6: 35 33 34 36	Day 5 average 27.50°C
Day 7: 30 31 32 30	Day 6 average 34.50°C
	Day 7 average 30.75°C
	Hottest Day: Day 6 (34.50°C)

Problem 14: Movie Hall Seat Reservation Tracker

A movie hall has n rows and m columns of seats.

Each seat can be either booked (1) or available (0).

Write a C program that:

1. Counts the total available seats.

2. Displays the row with the most booked seats.

Input: 5 6 1 0 1 1 0 1 0 0 1 0 0 1 1 1 1 1 1 1 0 1 0 0 1 0 1 0 0 1 0 0	Output: Total available seats = 15 Row 3 has the most booked seats: 6
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Additional Practice & Resources

To strengthen your understanding of loops and nested loop, practice more examples and exercises from the following trusted learning resource:

Online Resource:

W3School: <https://www.w3resource.com/c-programming-exercises/array/index.php>

Code for win: <https://www.w3resource.com/c-programming-exercises/while-loop/index.php>